

1. Which of the following components is used to create an N-type semiconductor?  
 a. Trivalent impurity  
 b. Pentavalent impurity  
 c. Intrinsic semiconductor  
 d. Extrinsic semiconductor  
 (a) a and c  
 (b) b and d  
 (c) b and c  
 (d) a and d [c]
2. The relationship between the magnetic  $\chi$  susceptibility and the magnetic permeability ( $\mu$ ) is given by:  
 ( $\mu_0$  is the permeability of free space and ( $\mu_r$  is relative permeability)  
 (a)  $\chi = \frac{\mu}{\mu_0} - 1$   
 (b)  $\chi = \frac{\mu_r}{\mu_0} + 1$   
 (c)  $\chi = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$   
 (d)  $\chi = 1 - \frac{\mu}{\mu_0}$  [a]
3. The first law of thermodynamics is based on the law of conservation of  
 (a) energy  
 (b) mass  
 (c) momentum  
 (d) None of these. [a]
4. Three capacitors each of capacity 4  $\mu\text{F}$  are to be connected in such a way that the effective capacitance is 6  $\mu\text{F}$ . This can be done by  
 (a) connecting all of them in series  
 (b) connecting them in parallel  
 (c) connecting two in series and one in parallel  
 (d) connecting two in parallel and one in series. [c]
5. When a charged particle enters into a magnetic field with velocity perpendicular to the direction of the magnetic field. The particle follows a curved path. What will happen to the radius of this curved path if the magnetic field is increased?  
 (a) particle goes in straight line.  
 (b) radius will remain same  
 (c) radius will increase  
 (d) radius will decrease [d]
6. If a velocity has both perpendicular and parallel components while moving through a magnetic field, what is the path followed by a charged particle?  
 (a) Circular  
 (b) elliptical  
 (c) Linear  
 (d) Helical [d]
7. The dimensional formula of constant  $a$  in van der Waals gas equation  $(p + \frac{a}{v^2})(v - b) = RT$ :  
 (a)  $[\text{ML}^3\text{T}^{-2}]$   
 (b)  $[\text{ML}^5\text{ST}^{-2}]$   
 (c)  $[\text{ML}^2\text{T}^{-1}]$   
 (d)  $[\text{ML}^2\text{T}^{-5}]$  [b]
8. The disintegration constant  $\lambda$  of a radioactive element is 0.00231 per day. Its half life is  
 (a) 500 days  
 (b) 400 days  
 (c) 200 days  
 (d) 300 days [d]
9. In  $\vec{E}$  and  $\vec{k}$  represent electric field and propagation vectors of the EM waves in vacuum, then magnetic field vector is given by:  
 ( $\omega$  - angular frequency):  
 (a)  $\omega(\vec{E} \times \vec{K})$   
 (b)  $\omega(\vec{K} \times \vec{E})$   
 (c)  $\vec{E} \times \vec{K}$   
 (d)  $\frac{1}{\omega}(\vec{K} \times \vec{E})$  [d]
10. Moment of inertia of a uniform circular disc about a diameter is  $I$ . Its moment of inertia about an axis perpendicular to its plane and passing through a point on its rim will be  
 (a) 4I  
 (b) 6I  
 (c) 3I  
 (d) 5I [b]
11. The ratio of the distances travelled by a freely falling body in the 1<sup>st</sup>, 2<sup>nd</sup>, 3<sup>rd</sup>, and 4<sup>th</sup>, second:  
 (a) 1:1:1:1  
 (b) 1:2:3:4  
 (c) 1:4:9:16  
 (d) 1:3:5:7 [d]
12. Given the current  $i(t)$  and voltage  $v(t)$  of a circuit element, the power  $p(t)$  and energy  $w(t)$  are given by:  
 (a)  $p(t) = v(t) i(t)$  and  $w(t) = \int_0^t i(\tau) d\tau$   
 (b)  $p(t) = v(t) i(t)$  and  $w(t) = \int_0^t p(\tau) d\tau$   
 (c)  $p(t) = v(t) i(t)$  and  $w(t) = \int_0^t v(\tau) d\tau$   
 (d)  $p(t) = v(t) + i(t)$  and  $w(t) = \int_0^t p(\tau) d\tau$  [b]
13. If the magnification produced by a lens is negative, then the image is:  
 (a) Virtual and erect  
 (b) Real and inverted  
 (c) Virtual and inverted  
 (d) Real and erect [b]

14. A measured temperature in Fahrenheit scale is  $200^{\circ}\text{F}$ . What will be its value in Celsius scale?  
(a)  $93.3^{\circ}\text{C}$   
(b)  $40^{\circ}\text{C}$   
(c)  $99^{\circ}\text{C}$   
(d)  $30^{\circ}\text{C}$  [a]
15. An athlete with a mass 60 kg in the Olympic Games covers a distance of 100 m in 10s. His kinetic energy can be estimated to be in the range of:  
(a) 200 J-500 J  
(b) 200000 J-500000 J  
(c) 20000 J-50000 J  
(d) 2000 J-5000 J [d]
16. If two resistors of  $10\text{ k}\Omega$  and  $30\text{ k}\Omega$  are connected in parallel what will be the value of the effective resistance (in  $\text{k}\Omega$ )?  
(a) 10  
(b) 12.5  
(c) 7.5  
(d) 5 [c]
17. The radius of curvature of a spherical mirror is 15. Find its focal length.  
(a) 9 cm  
(b) 8.5 cm  
(c) 7.5 cm  
(d) 2.7 cm [c]
18. A bus starts from rest and rolls down a hill with a constant acceleration of  $0.5\text{ m/s}^2$ . What is the force acting on it if its mass is 8 tonnes?  
(a) 40N  
(b) 400 N  
(c) 4000 N  
(d) 40000 N [c]
19. An object is placed at 10 cm in front of a concave mirror of radius of curvature 25 cm. Find the position of the image (in cm)?  
(a) -50  
(b) 40  
(c) 50  
(d) -40 [c]
20. 2 An unbalanced force of 90 N acts on an object and produces an acceleration of  $15\text{ m/s}^2$  in it. Then, the mass of the object is  
(a) 15 kg  
(b) 90 kg  
(c) 1350 kg  
(d) 6 kg [d]
21. A non-SI unit called 'nit' is the unit of which of the following photometric quantities used to measure a multitude of light intensity?  
(a) Luminous exposure  
(b) Luminance  
(c) Luminosity  
(d) Luminous emittance [b]
22. Dimensional formula of Pressure?  
(a)  $\text{ML}^0\text{T}^2$   
(b)  $\text{ML}^{-1}\text{T}^{-2}$   
(c)  $\text{MLT}^2$   
(d)  $\text{ML}^2\text{T}^{-1}$  [b]
23. The following is the relation between the object distance and the image distance in a plane mirror image :  
(a) object distance > image distance  
(b) object distance = image distance  
(c) object distance < image distance  
(d) not related [b]
24. Which trend is found about the distance between the galaxies?  
(a) Constant  
(b) Increasing  
(c) Unpredictable  
(d) Decreasing [b]
25. A ball possesses 3000 units of momentum. What would be the ball's new momentum if its velocity was doubled?  
(a) 3000 units  
(b) 6000 units  
(c) 20000 units  
(d) 9000 units [b]
26. The value of  $\int \frac{\log x}{(x+1)^2} dx$  is  
(a)  $\frac{-\log x}{x+1} + \log x - \log(x+1)$   
(b)  $\frac{\log x}{x+1} + \log x - \log(x+1)$   
(c)  $\frac{\log x}{x+1} \log x - \log(x+1)$   
(d)  $\frac{-\log x}{x+1} - \log x - \log(x+1)$  [a]
27.  $\begin{vmatrix} 1 & a & a^2 - bc \\ 1 & b & b^2 - ac \\ 1 & c & c^2 - ab \end{vmatrix}$  is equal to ?  
(a) 0  
(b) 3abc  
(c)  $(a+b+c)^3$   
(d)  $a^3 + b^3 + c^3 - 3abc$  [a]
28. The ratio in which the plane  $x - 2y + 3z = 17$  divides the line joining the points (-2, 4, 7) and (3, -5, 8) is  
(a) 10 : 3  
(b) 3 : 1  
(c) 3 : 10  
(d) 10 : 1 [c]

29. If A is a singular matrix, then  $A[\text{adj}(A)] = ?$   
 (a) A  
 (b)  $\text{adj}(A)$   
 (c)  $A^{-1}$   
 (d) Null matrix. [d]
30. If  $5 \tan A = 3$ , then the value of  $\frac{5 \sin A - 2 \cos A}{\sin A + 2 \cos A}$  is [d]  
 (a)  $7/9$   
 (b)  $5/13$   
 (c)  $9/13$   
 (d)  $5/7$  [b]
31.  $\int \frac{dx}{\sin^2 x \cos^2 x}$  equals :  
 (a)  $\tan x \cdot \cot x + c$   
 (b)  $\tan x - \cot x + c$   
 (c)  $\tan x - \cot 2x + c$   
 (d)  $\tan x \cot x + c$  [b]
32. The area of the region bounded by the parabola  $y^2 = 4ax$  and its latus rectum is:  
 (a)  $\frac{4a^2}{3}$  sq. units  
 (b)  $\frac{8a^2}{3}$  sq. units  
 (c)  $\frac{2a^2}{3}$  sq. units  
 (d)  $\frac{9a^2}{5}$  sq. units [b]
33. A biased coin with the probability of getting head equal to  $\frac{1}{4}$  is tossed five times. What is the probability of getting tail in all the first four tosses followed by head?  
 (a)  $\frac{81}{512}$   
 (b)  $\frac{81}{1024}$   
 (c)  $\frac{81}{256}$   
 (d)  $\frac{27}{1027}$  [b]
34. if  $y = p \cos 2x + q \sin 2x$ , then  $\frac{d^2 y}{dx^2}$  is equal to?  
 (a)  $-2y$   
 (b)  $-4y$   
 (c)  $2y$   
 (d)  $4y$  [b]
35. The distance between the parallel lines  $y - 8 = 0$  and  $y + 1 = 0$  is \_\_\_\_\_.  
 (a) 8  
 (b) 7  
 (c) 1  
 (d) 9 [d]
36. If  $\tan^{-1} k + \tan^{-1} \frac{1}{2} = \frac{\pi}{4}$ , then what is the value of K?  
 (a) 1  
 (b)  $1/2$   
 (c)  $1/3$   
 (d)  $1/4$  [c]
37. Solution of the differential equation  $\frac{dy}{dx} + \frac{y}{x} = \sin x$  is:  
 (a)  $x(y + \cos x) = \sin x + c$   
 (b)  $x(y - \cos x) = \sin x + c$   
 (c)  $xy \cos x = \sin x + c$   
 (d)  $x(y + \cos x) = \cos x + c$  [a]
38. Find the standard deviation of 8, 12, 13, 15, 22.  
 (a) 3.54  
 (b) 3.72  
 (c) 4.21  
 (d) 4.6 [d]
39. The volume of spherical ball is increasing at the rate of  $4\pi \text{cc/sec}$ . Then rate of surface area when volume is  $288\pi$   
 (a)  $\frac{3\pi}{4} \text{cm}^2/\text{sec}$   
 (b)  $\frac{4}{3} \text{cm}^2/\text{sec}$   
 (c)  $\frac{4\pi}{3} \text{cm}^2/\text{sec}$   
 (d)  $\frac{3}{4} \text{cm}^2/\text{sec}$  [c]
40. If  $A = \{1, 3, 6\}$  and  $B = \{5, 6\}$  then  $(A \cup B)$  will be:  
 (a)  $\{1, 2, 3, 4, 5\}$   
 (b)  $\{2, 4\}$   
 (c)  $\{1, 3, 5, 6\}$   
 (d)  $\{2, 4, 5\}$  [c]
41. Find the modulus of the complex number  $\frac{1+2i}{1-(1-i)}$   
 (a)  $\sqrt{5}$   
 (b)  $\sqrt{2}$   
 (c)  $\sqrt{7}$   
 (d) 1 [a]
42. The solution  $\frac{dx}{dt} = 3x + 8$  will be  
 (a)  $x = \frac{1}{3}e(t+c) - \frac{8}{3}$   
 (b)  $x = \frac{1}{3}e^3(t+c) - \frac{8}{3}$   
 (c)  $x = \frac{1}{3}e(t+c) + \frac{8}{3}$   
 (d)  $x = \frac{1}{3}e(t+c) + \frac{8}{3}$  [b]
43. In how many ways can 8 persons sit in a row?  
 (a)  $7!$   
 (b)  $8!$   
 (c)  $6!$   
 (d) None of these [b]
44. For any  $n \in \mathbb{N}$ ,  $\frac{1}{2.5} + \frac{1}{5.8} + \dots + \frac{1}{(3n-1)(3n+2)} =$   
 (a)  $\frac{n}{6n+4}$   
 (b)  $\frac{n^2}{6n+4}$   
 (c)  $\frac{1}{2} \frac{n^2}{6n+4}$   
 (d)  $\frac{n}{6n^2+4}$  [a]

45. What is the distance between the lines  $3x + 4y = 9$  and  $6x + 8y = 18$ ?  
 (a) 0  
 (b) 3 units  
 (c) 9 units  
 (d) 18 units [a]
46. The 6<sup>th</sup> term of a G.P. is 32 and its 8<sup>th</sup> term is 128, then the common ratio of the G.P. is  
 (a) -1  
 (b) 2  
 (c) 4  
 (d) -4 [b]
47. The middle term in the expansion of  $\left(x^2 + \frac{1}{x^2} + 2\right)^n$  is  
 (a)  $\frac{(2n)!}{(n!)^2}$   
 (b)  $\frac{(2n)!}{\left[\left(\frac{n}{2}\right)!\right]^2}$   
 (c)  $\frac{n}{\left[\left(\frac{n}{2}\right)!\right]^2}$   
 (d)  $\frac{1.2.5...(2n+1)}{n!} 2^n$  [a]
48. The function  $f(x) = x^3 - 3x^2 + 6$  is an increasing function for:  
 (a)  $0 < x < 2$   
 (b)  $x < 2$   
 (c)  $x > 2$  or  $x < 0$   
 (d) all x [c]
49. The domain of  $\sin^{-1}(x)$  is  
 (a)  $(-\pi, \pi)$   
 (b)  $[-1, 1]$   
 (c)  $(0, 2\pi)$   
 (d)  $(-\infty, \infty)$  [b]
50. Find the area bounded between the curve  $y = x$  and  $y = x^2$ .  
 (a)  $1/2$   
 (b)  $1/4$   
 (c) 1  
 (d)  $1/6$  [d]
51. The following sentence has been split into four segments. Identify the segment that contains a grammatical error.  
**I love the dogs /; but I don't like / the dog my friend Tintie / owns since it bites people.**  
 (a) owns since it bites people  
 (b) ; but I don't like  
 (c) I love the dogs  
 (d) the dog my friend Tintie [c]
52. The following sentence has been split into four segments. Identify the segment that contains a grammatical error.

- Are you sure that / you will be broadcast/ the news to everyone / across the country?  
 (a) the news to everyone  
 (b) across the country  
 (c) you will be broadcast  
 (d) Are you sure that [c]
53. Direction: The following sentence has been split into four segments. Identify the segment that contains a grammatical error.  
**Don't let the failure / demotivate you, as / failures are the / stepping stones to success.**  
 (a) failures are the  
 (b) Don't let the failure  
 (c) stepping stones to success  
 (d) demotivate you as [b]
54. Select the option that can be used as a one-word substitute for the given group of words.  
**A sudden and violent change or upheaval**  
 (a) Tranquility  
 (b) Serenity  
 (c) Equanimity  
 (d) Turbulence [d]
55. Select the option that can be used as a one-word substitute for the given group of words.  
**The place where soldiers live**  
 (a) Unit  
 (b) Regiment  
 (c) Brigade  
 (d) Barracks [d]
56. Select the option that can be used as a one-word substitute for the given group of words.  
**one who does something for pleasure**  
 (a) Anarchist  
 (b) Atheist  
 (c) Hedonist  
 (d) Aggressor [c]
57. Select the option that expresses the given sentence in passive voice.  
**Rover would not swallow the pill.**  
 (a) The pill would not be swallowing by Rover.  
 (b) The pill would not swallowed by Rover.  
 (c) The pill would not be swallowed by Rover.  
 (d) The pill would not be swallow by Rover. [c]
58. Select the option that expresses the given sentence in passive voice.  
**Does Sneha not lick the butter?**  
 (a) Is the butter being licked by Sneha?  
 (b) Is the butter not licked by Sneha?  
 (c) Sneha does not lick the butter?  
 (d) Was Sneha not licking the butter? [b]

**59. Select the option that expresses the given sentence in passive voice.**

**The magazine house was expecting unpublished authors for their forthcoming exclusive issue on new trends.**

- (a) Unpublished authors will be expected by the magazine house for their forthcoming exclusive issue on new trends.
- (b) Unpublished authors were being expected by the magazine house for their forthcoming exclusive issue on new trends.
- (c) Unpublished authors was expected by the magazine house for their forthcoming exclusive issue on new trends.
- (d) Unpublished authors were expected by the magazine house for their forthcoming exclusive issue on new trends. **[b]**

**60. Select the correct indirect form of the given sentence.**

**Ramita said, "I wish I didn't have to meet my step-mother."**

- (a) Ramita said she wished she hadn't have to meet her step-mother.
- (b) Ramita said she wished she didn't meet her step-mother.
- (c) Ramita said she wished she didn't have to meet her step-mother.
- (d) Ramita said she wished hadn't met her step-mother. **[c]**

**61. Select the correct indirect form of the given sentence.**

**He says to Gayatri, "You are writing an apology to the warden."**

- (a) He told Gayatri that she is writing an apology to the warden.
- (b) He tells Gayatri that she is writing an apology to the warden.
- (c) He says to Gayatri that she was writing an apology to the warden.
- (d) He tells Gayatri that she was writing an apology to the warden. **[b]**

**62. Select the INCORRECTLY spelt word.**

- (a) Practice
- (b) Quarentine
- (c) Concentrate
- (d) Elicit **[b]**

**63. Select the INCORRECTLY spelt word.**

- (a) Numerous
- (b) Critiqued
- (c) Inacuracies
- (d) Inconsistencies **[c]**

**64. Select the INCORRECTLY spelt word.**

- (a) Analytically
- (b) Appropriateately
- (c) Apparently
- (d) Astonishingly **[b]**

**65. Select the INCORRECTLY spelt word.**

- (a) Renewel
- (b) Horizon
- (c) Composer
- (d) Algebra **[a]**

**66. Read the following passage about mother tongue and answer the questions that follow.**

Formal education through the medium of a language that is not a child's mother tongue, leading children not learning their own language fully (not learning the language that the parent generation knows) This is a subtractive language learning situation, where the dominant language is learned at the cost of the mother tongue. Instead, knowledge in dominant languages can be added to the children's linguistic reperto at no cost to the mother tongues. In this kind of additive language learning situation the result is additive bilingualism or multilingualism, and the maintenance of the mother tongue. Subtractive language learning also means that these children will tend not to speak their mother tongue to their own children.

Homogenizing mass media, entertainment and other cultural products in dominant languages takes place through which these products send the message that the languages in which films, TV and pop music come have a higher status than others. Urbanization, migration and employment mobility of people often lead to disintegration of language communities, with fewer chances for children to hear and use their parents' language(s) on a daily basis. A labor market which requires knowledge of dominant languages and does no offer economic or psychological incentives for maintaining the smaller languages. Insufficient protection c linguistic human rights and ideologies see not multilingualism but monolinguals in dominant languages as normal, sufficient and desirable, both for states (one nation one language) and for individuals. This leads parents often thinking that their children have to choose between either learning their mother tongue (and losing out on the labor market) or learning the dominant language (and sacrificing their own language).

**What is known as the 'additive language learning situation'?**

- (a) Learning through English medium.
  - (b) Learning another language without harming the mother tongue.
  - (c) Learning through mother tongue
  - (d) Learning mother tongue with the support of another language.
- [b]**

**67. Choose the correct statement.**

In the current scenario of language learning:

- (a) Parents have no choice but to choose between mother tongue or dominant language.
  - (b) Parents have no choice but do not choose mother tongue.
  - (c) Parents have choice, but choose the dominant language to the mother tongue
  - (d) Government has no choice, but to impose a dominant language among its people.
- [c]**

**68. Which of the following statements, according to the author, is TRUE?**

- (a) Homogenisation of certain instruments of society leads to language hierarchies.
  - (b) Mother tongue is always the dominant language in society.
  - (c) Some language are superior and some are inferior,
  - (d) Language domination takes place through political interference.
- [a]**

**69. What is 'subtractive language learning'?**

- (a) Learning to subtract in languages
  - (b) Learning the dominant language which undermines the mother tongue
  - (c) Learning English at the cost of mother tongue
  - (d) Simultaneously learning two languages which support each other
- [b]**

**70. The following sentence has been split into four segments. Identify the segment that contains a grammatical error.**

**I noticed that they/ did not went to / the museum but entered / a shopping complex.**

- (a) the museum but entered
  - (b) a shopping complex.
  - (c) did not went to
  - (d) I noticed that they
- [c]**